ELEG 3143 Assignment # 10

1. For each of the following joint pdf, identify whether the random variables are independent, and find $\mathbb{E}[XY]$.

   (a)

   $$f_{XY}(x,y) = \begin{cases} \frac{kx}{y}, & 0 \le x \le 1, 1 \le y \le 2 \\ 0, & \text{otherwise} \end{cases} \tag{1}$$

   (b)

   $$f_{XY}(x,y) = \begin{cases} k(xy + 2x + 3y + 6), & 0 \le x \le 1, 0 \le y \le 1 \\ 0, & \text{otherwise} \end{cases} \tag{2}$$

2. Consider two independent Gaussian RVs with means 1 and 2, respectively, and variances 1 and 4, respectively.Find the probability that $XY > 0$.

3. Two RVs have zero mean and variances 16 and 36, respectively. Their correlation coefficient is 0.5. Find the variance of their sum.

4. Two independent RVs, X and Y, have variances $\sigma_X^2 = 9$ and $\sigma_Y^2 = 25$. Two new RVs are defined as

   $$U = 3X + 4Y \tag{3}$$
   $$V = 5X - 2Y \tag{4}$$

   (a) Find the variances of U and V.

   (b) Find the correlation coefficient of U and V.

5. Consider two independent RVs X and Y with variances $\sigma_X^2 = 9$ and $\sigma_Y^2 = 25$. Define $Z = X + Y$.

(a) Find the correlation coefficient between X and Y.

(b) Find the correlation coefficient between Y and Z.

(c) Find the variance of Z.

6. Consider two independent RVs X and Y. The pdf of X is

$$f_X(x) = \begin{cases} 2x, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases} \tag{5}$$

The RV Y is uniformly distributed between -1 and 1. Find the pdf of $Z = X + 2Y$.

7. Consider two independent RVs X and Y with pdf given as

$$f_X(x) = 5e^{-5x}u(x) \tag{6}$$
$$f_Y(y) = 2e^{-2y}u(y) \tag{7}$$

Define $Z = X + Y$.

(a) Find the pdf of Z.

(b) Find the probability that Z is greater than 0.1.

8. Consider two random variables X and Y with joint pdf

$$f_{X,Y}(x, y) = 4xy, 0 < x < 1, 0 < y < 1.$$

Define two new random variables $Z = X + Y$ and $W = X - y$. Find the joint pdf of Z and W.